

The Impact of Weather Extremes on Firm-Level Productivity

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- Climate extremes are increasing in frequency and intensity (IPCC, 2023)
- Macro evidence links temperature to GDP (Burke et al., 2015; Kotz et al., 2021)
- But firm-level mechanisms remain poorly understood:
 - Most studies focus on single countries or single weather dimensions
 - Heat stress dominates the literature; cold extremes, variability largely ignored
- **Gap:** No large-scale, multi-country, multi-weather-dimension firm-level evidence

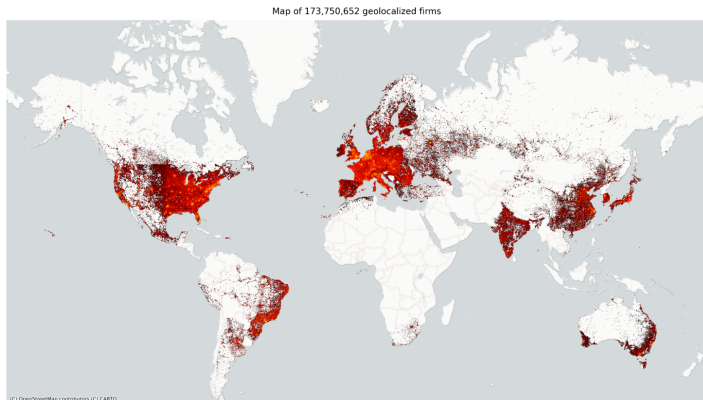
What we do:

- Match **20M+ firm-year observations** (ORBIS) with high-resolution weather data (ERA5-Land) across **38 countries**
- Estimate effects of multiple weather dimensions on turnover per employee
- Decompose heterogeneity by Köppen climate zones

Key findings:

- **Heatwaves:** -0.25% labor productivity per σ increase
- **Coldwaves:** -0.19% per σ increase
- Heatwaves harmful across *all* climate zones
- Temperature variability significant only in temperate climates

- **ORBIS** database (Bureau van Dijk, Oct 2024)
- Balance sheet + income statement data, 2000–2022
- Firms with 5–250 employees, ≥ 5 years of data
- Three-stage geocoding (street $\rightarrow$ postal code $\rightarrow$ BvD coordinates)
- DV: Log(turnover per employee)
- 20.5M firm-year observations



541,618 unique firm locations

Source: ERA5-Land reanalysis (ECMWF), 0.1° grid, daily

Variable	Definition
Mean Temperature (TM)	Annual mean of daily mean 2m temperature
Temp. Variability (TVAR)	RMSD of daily temp. vs. long-run daily mean
Heatwaves (HW)	Severity of worst event (≥ 5 days, $TX > TX_{90th} & 30^{\circ}C$)
Coldwaves (CW)	Severity of worst event (≥ 5 days, $TN < TN_{10th} & 0^{\circ}C$)
Extreme Rainfall	Largest 5-day cumulated precipitation
Extreme Snowfall (SNX)	Largest 3-day cumulated snowfall

All variables standardized $\Rightarrow$ coefficients = % change per σ

$$\ln(y_{i(gsc)t}) = \mathbf{W}'_{i(gsc)t}\beta + \mathbf{X}'_{i(gsc)t-1}\gamma + \delta_i + \zeta_{sct} + \varepsilon_{i(gsc)t}$$

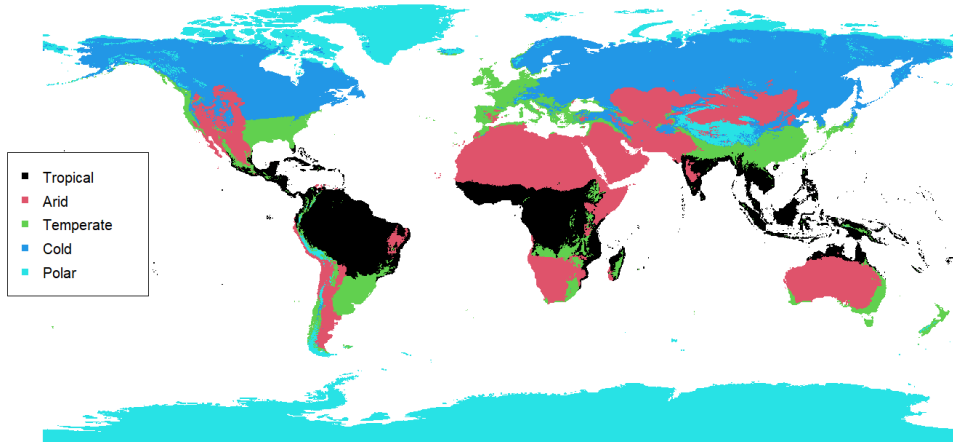
- y : turnover per employee
- $\mathbf{W}$: mean temp., temp.², temp. variability, heatwaves, coldwaves, extreme rainfall, extreme snowfall
- $\mathbf{X}$: lagged firm controls (size, age, profitability, capital intensity)
- δ_i : firm FE $\Rightarrow$ within-firm variation over time
- ζ_{sct} : country $\times$ sector $\times$ year FE
- SE clustered at grid-cell level

	Log(turnover per employee)
Mean Temperature	0.023**
(Mean Temperature) ²	−0.014*
Temperature Variability	−0.002
Heatwaves	−0.00246***
Coldwaves	−0.00193***
Extreme Rainfall	−0.000
Extreme Snowfall	0.001
N	20,546,849
R ²	0.912
Firm FE / Country-Sector-Year FE	Yes / Yes

* p<0.1, ** p<0.05, *** p<0.01. SE clustered by grid cell.

- Heatwaves: -0.25% per σ | Coldwaves: -0.19% per σ
- Temp. variability negative but *not* significant in full sample

Climate zones



	Temperate	Cold	Arid
Temp. Variability	−0.0062 ^{***}	0.003	0.004
Heatwaves	−0.0023 ^{***}	−0.0019 ^{**}	−0.0017 ^{**}
Coldwaves	−0.000	−0.0032 ^{***}	−0.0029 ^{***}
Extreme Snowfall	−0.0009 ^{**}	0.002	0.001
N	11.4M	7.7M	1.4M

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Key rows shown.

- Heatwaves harmful across all climate zones
- Temperature variability: significant only in temperate zones
- Coldwaves: concentrated in cold and arid zones
- Extreme snowfall: significant only in temperate zones

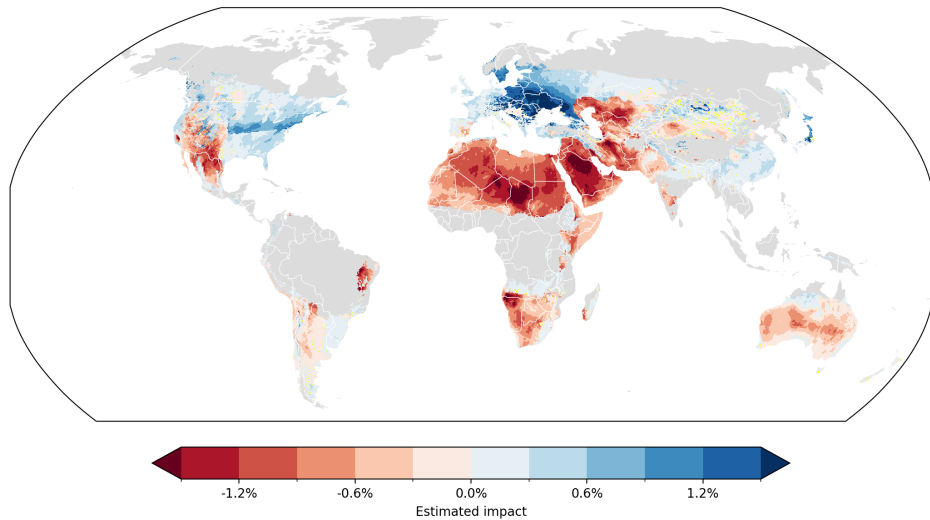
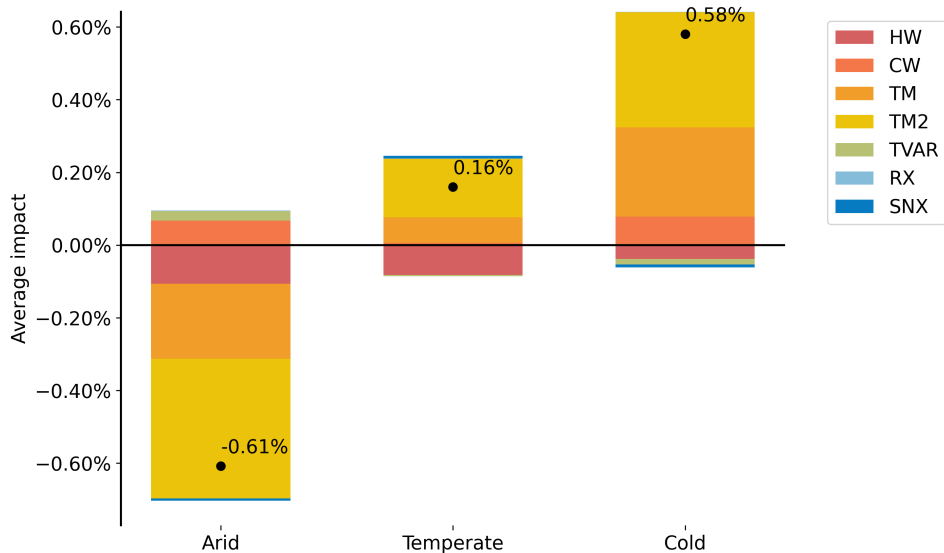


Figure 3. Difference 1980-1989 and 2010-2019



Key findings:

1. Inverted U-shape between mean temperature and productivity confirmed at firm level
2. **Heatwaves** (-0.25%) and **coldwaves** (-0.19%) are the dominant weather determinants
3. Climate zone heterogeneity: heatwaves universal, temp. variability localized to temperate zones

Next steps:

- Adding more firms
- Adding more extremes (LASSO from Akyapi et al. (2024))

Thank you!

Questions & comments welcome

References

Akyapi, B., Bellon, M., and Massetti, E. (2024). Estimating macro-fiscal effects of climate shocks from billions of geospatial weather observations. *American Economic Journal: Macroeconomics*.

Burke, M., Hsiang, S. M., and Miguel, E. (2015). Global non-linear effect of temperature on economic production. *Nature*, 527(7577):235–239.

Kotz, M., Wenz, L., Stechemesser, A., Kalkuhl, M., and Levermann, A. (2021). Day-to-day temperature variability reduces economic growth. *Nature Climate Change*, 11(4):319–325.

Backup Slides

Backup: Full Main Results

	Log(turnover per employee)		
Mean Temperature	0.02306** (0.00938)		
(Mean Temperature) ²	−0.01353* (0.00759)		
Temperature Variability	−0.00226 (0.00171)		
Heatwaves	−0.002455*** (0.000456)		
Coldwaves	−0.00193*** (0.00052)		
Extreme Rainfall	−0.000234 (0.000460)		
Extreme Snowfall	0.000828 (0.000548)		
N	20,546,849		
R ²	0.912	R ² Adj. = 0.902	RMSE = 0.40
Financial Controls	Yes		
Firm FE / Country-Sector-Year FE	Yes / Yes		

* p<0.1, ** p<0.05, *** p<0.01. SE clustered by grid cell in parentheses.

Backup: Full Climate Zone Results

	Temperate	Cold	Arid
Mean Temperature	0.00228 (0.01493)	0.00920 (0.00737)	−0.0045 (0.0188)
(Mean Temperature) ²	0.00463 (0.01559)	0.01430** (0.00685)	−0.00606 (0.01722)
Temperature Variability	−0.00615*** (0.00149)	0.00322 (0.00210)	0.00404 (0.00357)
Heatwaves	−0.002302*** (0.000433)	−0.001857** (0.000836)	−0.001742** (0.000787)
Coldwaves	−0.000147 (0.000362)	−0.003159*** (0.000898)	−0.002933*** (0.000716)
Extreme Rainfall	−0.000609 (0.000394)	0.000552 (0.001506)	0.000341 (0.000469)
Extreme Snowfall	−0.000853** (0.000423)	0.00167 (0.00113)	0.00104 (0.00155)
N	11,440,829	7,693,660	1,393,133
R ²	0.896	0.888	0.829
RMSE	0.31	0.52	0.32

* p<0.1, ** p<0.05, *** p<0.01. SE in parentheses.

Single regression 1980-1989 vs. 2010-2019

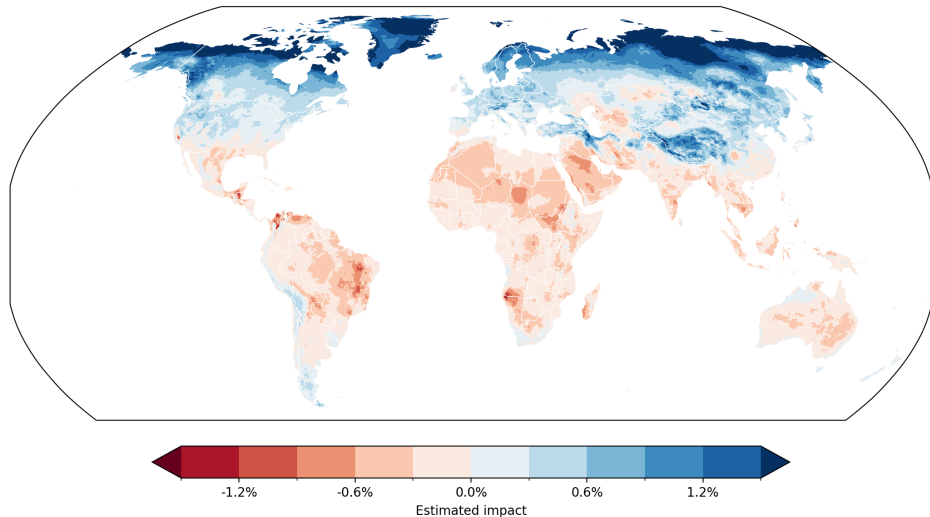


Figure 5. Aggregate regression difference 1980-1989 and 2010-2019

Backup: Adaptation Results

	Log(turnover per employee)
<i>Weather shocks</i>	
Mean Temperature Shock	0.0072** (0.0035)
Mean Temperature ² Shock	−0.0079** (0.0039)
Temp. Variability Shock	−0.0108*** (0.0011)
Rainfall Shock	−0.0019** (0.0009)
Heatwaves Shock	−0.0032*** (0.0003)
Coldwaves Shock	0.0001 (0.0004)
Snow Shock	−0.0005 (0.0006)
Drought Shock	−0.0007* (0.0004)
<i>Shock × Climate interactions</i>	
TVAR shock × TVAR climate	0.0040*** (0.0008)
RR shock × RR climate	0.0021*** (0.0005)
SNX shock × SNX climate	0.0008*** (0.0003)
HW shock × HW climate	0.0004 (0.0004)
CW shock × CW climate	0.0000 (0.0006)
N	11,607,314
R ²	0.803
Firm FE / Country-Sector-Year FE	Yes / Yes

* p<0.1, ** p<0.05, *** p<0.01. SE in parentheses.

Backup: Adaptation — Key Takeaway

Framework: Decompose weather into shock (deviation from long-run mean) and climate (long-run mean). Interact to test adaptation.

Evidence of adaptation:

- **Temperature variability:** firms in high-variability locations experience attenuated losses
- **Rainfall:** firms in wetter climates cope better with precipitation shocks
- **Snowfall:** firms in snow-prone regions less affected by snow shocks

No adaptation to extreme heat:

- HW shock $\times$ HW climate: small, insignificant
- Even firms in historically hot regions remain **equally vulnerable** to heatwave shocks
- Consistent with physiological limits to heat adaptation

Backup: Summary Statistics

	Min	P5	P25	Median	Mean	P75	P95	Max	SD
Mean Temperature	−16.0	3.7	8.6	12.2	11.6	15.1	17.7	29.4	4.5
Temp. Variability	0.55	1.83	2.34	2.79	2.91	3.41	4.43	7.30	0.79
Heatwaves	0.0	0.0	0.0	0.0	12.1	17.3	56.7	266.5	23.7
Coldwaves	0.0	0.0	0.0	0.0	11.4	13.3	62.0	285.0	23.2
Extreme Rainfall	1.03	36.17	49.76	67.08	82.01	98.11	182.59	765.08	48.28
Extreme Snowfall	0.00	0.00	3.13	11.78	13.40	19.83	35.57	234.93	12.56
Num. Employees	5	5	8	15	29	33	109	249	37

Backup: Sample Coverage by Country

Country	N
ES	3,506,349
RU	3,156,216
IT	2,891,460
FR	1,420,176
JP	1,346,199
UA	1,106,633
PT	942,626
SE	696,247
RO	639,445
HU	538,311
BG	505,719
KR	429,858
GB	386,099
HR	320,337

Country	N
RS	297,392
PL	295,057
FI	283,657
NO	258,836
BE	257,008
GR	237,566
CN	196,775
CZ	186,772
LV	131,510
EE	123,904
SK	112,595
MK	73,457
LT	61,194
DE	58,243

Country	N
DK	40,495
NL	13,847
ME	13,473
IE	10,760
IS	7,380
AT	2,073
CH	1,272
LU	1,195
AU	849
MT	296
MX	204
TR	62
BR	43
VN	8

42 countries